

Iron intake and the risk of type 2 diabetes in women: a prospective cohort study.

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OBJECTIVE: Epidemiological studies suggest that high body iron stores are associated with insulin resistance and type 2 diabetes. The aim of this study was to evaluate the association between dietary intake of iron and the risk of type 2 diabetes. RESEARCH DESIGN AND METHODS: We conducted a prospective cohort study within the Nurses' Health Study. We followed 85,031 healthy women aged 34-59 years from 1980 to 2000. Dietary data were collected every 4 years, and data on medical history and lifestyle factors were updated biennially. RESULTS: During the 20 years of follow-up, we documented 4,599 incident cases of type 2 diabetes. We found no association between total, dietary, supplemental, or nonheme iron and the risk of type 2 diabetes. However, heme iron intake (derived from animal products) was positively associated with risk; relative risks (RRs) across increasing quintiles of cumulative intake were 1.00, 1.08 (95% CI 0.97-1.19), 1.20 (1.09-1.33), 1.27 (1.14-1.41), and 1.28 (1.14-1.45) (P(trend) < 0.0001) after controlling for age, BMI, and other nondietary and dietary risk factors. In addition, when we modeled heme iron in seven categories, the multivariate RR comparing women who consumed ≥ 2.25 mg/day and those with intake < 0.75 mg/day was 1.52 (1.22-1.88). The association between heme iron and the risk of diabetes was significant in both overweight and lean women. CONCLUSIONS: This large cohort study suggests that higher heme iron intake is associated with a significantly increased risk of type 2 diabetes.